Practical Session 5: LLM-Assisted Business Valuation

Hands-On: FCFF Computation, CoT Forecasting, and Monte Carlo

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Session Overview

Duration: 1 hour

1. **Recap** (5 min) — FCFF formula, DCF equation, error propagation, RAGAS metrics
2. **Problem 1** (15 min) — Compute FCFF and DCF enterprise value from raw financials
3. **Problem 2** (10 min) — Design a CoT forecasting prompt for a given company
4. **Case Study** (20 min) — Run a Monte Carlo DCF; evaluate sensitivity to WACC and g; answer three diagnostic questions
5. **Discussion** (5 min) — When is the DCF estimate unreliable? How would you report uncertainty?
6. **Solutions** (5 min) — Walkthrough of Problems 1 and 2

What you will practise

Translating raw financial statement data into FCFF; structuring CoT forecasting prompts; parameterising and interpreting a Monte Carlo DCF;

Recap: FCFF and DCF

Free Cash Flow to Firm:

$$\mathrm{FCFF}_t = \mathrm{EBIT}_t(1-\tau) + \mathrm{DA}_t - \mathrm{CapEx}_t - \Delta\mathrm{NWC}_t$$

DCF enterprise value:

$$V = \sum_{t=1}^{n} \frac{\mathrm{FCF}_t}{(1+\mathrm{WACC})^t} + \frac{\mathrm{FCF}_{n+1}}{(\mathrm{WACC}-g)(1+\mathrm{WACC})^n}$$

Relationship FCFF → FCFE:

$$\mathrm{FCFE}_t = \mathrm{FCFF}_t - \mathrm{InterestExp}_t(1-\tau) + \mathrm{NB}_t$$

Error propagation (key result):

$$\frac{\partial V}{\partial g} = \frac{\mathrm{FCF}_n(1+\mathrm{WACC})}{(\mathrm{WACC}-g)^2}$$

At $\mathrm{WACC} = 9\%$, $g = 3\%$: a 100-bps error in $g \to$ **15–18% error in EV.**

Recap: CoT Forecasting and Evaluation Metrics

CoT four-stage structure:

1. Macro context (interest rates, GDP, inflation)
2. Industry dynamics (drivers, headwinds, competitive dynamics)
3. Firm-specific (competitive position, CapEx plans, margin trajectory from MD&A)
4. Synthesis: point estimate + P10 downside + P90 upside per year

Three evaluation metrics:

Metric	Formula	Finance use case
MAPE	$\frac{1}{n}\sum \lvert y_t - \hat{y}_t \rvert / \lvert y_t \rvert$	Magnitude accuracy
RMSE	$\sqrt{\frac{1}{n}\sum (y_t - \hat{y}_t)^2}$	Sensitivity to outliers
DA	$\frac{1}{n}\sum \mathbf{1}[\text{sign match}]$	Investment decisions

Always compare against random-walk baseline and analyst consensus.

Problem 1: Compute FCFF and Enterprise Value (15 min)

A mid-cap industrial company reports the following for fiscal year 2023 (USD millions):

Revenue	2,850
EBIT	428
Depreciation & Amortisation	195
Capital Expenditure	240
NWC (FY2023)	380
NWC (FY2022)	315
Income Tax Expense	85
EBT (Earnings Before Tax)	368
Interest Expense	60
Net New Borrowing	$+40$

Your tasks:

1. Compute FCFF using the effective tax rate (infer from income tax / EBT).
2. Compute FCFE.

Problem 2: Design a CoT Forecasting Prompt (10 min)

You are building a valuation agent for the same industrial company. The MD&A states:

> *"We expect continued demand in our infrastructure segment driven by public investment in grid modernisation. However, supply chain bottlenecks in copper and steel may compress margins in the near term. We are targeting a 200-basis-point EBIT margin improvement over three years through operational efficiency programs. CapEx will remain elevated at 8–9% of revenue as we expand two manufacturing facilities."*

Your tasks:

1. Write the four-stage CoT reasoning (macro → industry → firm → synthesis) that a well-designed prompt should elicit.
2. Identify at least one potential internal inconsistency in the MD&A guidance that a consistency-checking step should flag.
3. Specify the JSON output format for Year 1–3 FCF forecasts (base, upside, downside, justification).

Case Study: Monte Carlo DCF (20 min)

Setup: You will parameterise and run a Monte Carlo DCF simulation using the FCFF computed in Problem 1 as the base (FCFF_0).

Parameters:

Parameter	Mean	Std Dev
WACC	10%	1.5%
Explicit growth (per year)	6%	3%
Terminal growth	2.5%	1%

Simulation spec:

- $M = 5{,}000$ draws; 5-year explicit horizon; `seed = 42`
- Clip terminal growth so $g_{\text{terminal}} < \text{WACC} - 0.5\%$
- Report: mean, median, std, P10, P25, P75, P90 of EV distribution

Three diagnostic questions on next slides.

Case Study Step 1: Run the Simulation

```
import numpy as np

def monte_carlo_dcf(fcff0, g_exp_mean, g_exp_std, wacc_mean,
    wacc_std,
                    g_term_mean, g_term_std, n=5, M=5000,
                        seed=42):
    rng = np.random.default_rng(seed)
    wacc = rng.normal(wacc_mean, wacc_std, M)
    g_term = rng.normal(g_term_mean, g_term_std, M)
    g_exp = rng.normal(g_exp_mean, g_exp_std, (M, n))

    # Clip terminal growth
    g_term = np.minimum(g_term, wacc - 0.005)

    ev = np.zeros(M)
    for m in range(M):
        fcfs = [fcff0]
        for t in range(n):
            fcfs.append(fcfs[-1] * (1 + g_exp[m, t]))
        pv_fcf = sum(fcfs[t]/(1+wacc[m])**t for t in range
            (1, n+1))
```

Case Study: Three Diagnostic Questions

After running the simulation, answer:

Question A: Compare the P10 and P90 enterprise values. What fraction of the total uncertainty in EV is attributable to terminal value uncertainty vs. explicit-period FCF uncertainty? (Hint: re-run with $\sigma_g = 0$ to isolate explicit-period uncertainty.)

Question B: At what terminal growth rate g does the median EV approximately double relative to the base-case DCF computed in Problem 1? (Hint: solve $TV \propto 1/(\mathrm{WACC} - g)$ for the target g.)

Question C: Suppose a competitor is acquired at 12× EV/EBITDA. The company's EBITDA is EBIT + DA = $623M. Does the comparables-based EV fall within the Monte Carlo P10–P90 range? What does the gap (or lack of gap) tell you about market pricing?

Case Study: Fill In This Table

Statistic	Value (USD M)
Mean EV	?
Median EV	?
Std Dev	?
P10 (bear)	?
P25	?
P75	?
P90 (bull)	?
P90 – P10 range	?
P90 / P10 ratio	?
Comparables EV (12× EBITDA)	?

Interpret: Is the DCF estimate consistent with the comparables estimate? What does a P90/P10 ratio > 2 suggest about this valuation?

Discussion: Communicating Valuation Uncertainty (5 min)

Discuss in pairs:

When is the DCF estimate unreliable?

- When $g \approx$ WACC (terminal value denominator approaches zero)?
- When the company has negative or volatile FCF (e.g., early-stage biotech)?
- When the forecast horizon is very short (terminal value dominates)?

How would you report uncertainty to a client?

- Point estimate only? (Never — why not?)
- Point estimate + range (P10–P90)?
- Full distribution histogram?
- Bull/base/bear narrative + Monte Carlo range?

Governance question: You run this pipeline on 200 companies daily. What HITL checkpoint(s) would you add, and at what threshold? (Hint: consider P90/P10 ratio, FCFF extraction confidence, and number of non-recurring items flagged.)

Solutions: Problem 1 — FCFF and DCF

Effective tax rate: $\tau = 85/368 = 23.1\%$

FCFF:

$$\begin{aligned} \Delta\text{NWC} &= 380 - 315 = 65 \text{ M} \\ \text{FCFF} &= 428(1 - 0.231) + 195 - 240 - 65 \\ &= 329.3 + 195 - 240 - 65 = \mathbf{219.3} \text{ M} \end{aligned}$$

FCFE:

$$\begin{aligned} \text{NI} &= (428 - 60)(1 - 0.231) = 368 \times 0.769 = 283.0 \text{ M} \\ \text{FCFE} &= 283.0 + 195 - 240 - 65 + 40 = \mathbf{213.0} \text{ M} \end{aligned}$$

Verify: $\text{FCFF} - \text{InterestExp}(1 - \tau) + \text{NB} = 219.3 - 60(0.769) + 40 = 219.3 - 46.1 + 40 = 213.2 \text{ M} \approx 213.0 \text{ M} \checkmark$

DCF: Using python_executor with `fcff0 = 219.3`, WACC $= 10\%$, $g_{\text{exp}} = 6\%$, $g_{\text{term}} = 2.5\%$, $n = 5$: **EV ≈ 3,672 M.**

Solutions: Problem 2 — CoT Prompt Key Points

Stage 3 (firm-specific) — key reasoning points:

- Positive: grid modernisation tailwind supports infrastructure segment revenue
- Negative: copper/steel supply chain bottlenecks compress near-term margins
- Positive: 200-bps margin improvement target over 3 years
- Negative: CapEx at 8–9% of revenue is elevated, limiting FCF conversion

Potential internal inconsistency to flag:

> *"200-bps EBIT margin improvement" while simultaneously "CapEx at 8–9% of revenue as we expand manufacturing facilities".*

Margin improvement is more plausible once the expansion CapEx phase ends. If the improvements are targeted during the high-CapEx period, a consistency-checking agent should flag this as a tension requiring clarification.

JSON schema: `{"year": 1, "base": <float>, "upside":`

Key Takeaways

1. **FCFF is not a data field in any filing.** It must be derived from EBIT, DA, CapEx, ΔNWC, and the effective tax rate. Each input can carry LLM extraction error; errors propagate multiplicatively.
2. **CoT separates narrative from number.** A good forecasting prompt forces the model to commit to a macro/industry/firm narrative *before* generating numbers. Inconsistencies become detectable.
3. **Terminal growth rate is the dominant uncertainty source.** A 100-bps error in g produces a 15–18% error in enterprise value. Always report Monte Carlo percentile ranges, not just point estimates.
4. **Comparables and DCF are complements, not substitutes.** A large gap between the two signals either market mispricing or a flawed assumption in one of the models. Investigate rather than average.
5. **Production pipelines need hallucination guards.** Validate every ticker; trace every financial figure to a specific accession number. A single hallucinated comparable can distort the entire multiple analysis.

Appendix

Stretch Problem: Sensitivity Decomposition

Extra / optional material — beyond the core lecture.

Stretch Problem: Decompose the EV Variance

For students who finish early, extend the Monte Carlo case study.

Goal: quantify how much of the variance in enterprise value is driven by each uncertain input (WACC, explicit growth, terminal growth).

Tasks:

1. Re-run the simulation three times, each time *freezing* two of the three inputs at their mean and letting only one vary.
2. Record $\mathrm{Var}(EV)$ for each single-input run; express each as a fraction of the all-inputs-varying variance.
3. **Predict, then check:** which input dominates? Reconcile with the analytic result $\frac{\partial V}{\partial g} = \frac{\mathrm{FCF}_n(1+\mathrm{WACC})}{(\mathrm{WACC}-g)^2}$.

Expected finding

Terminal-growth uncertainty should dominate, consistent with the 15–18% EV error per 100-bps g shock from the recap. The decomposition makes the analytic sensitivity tangible.